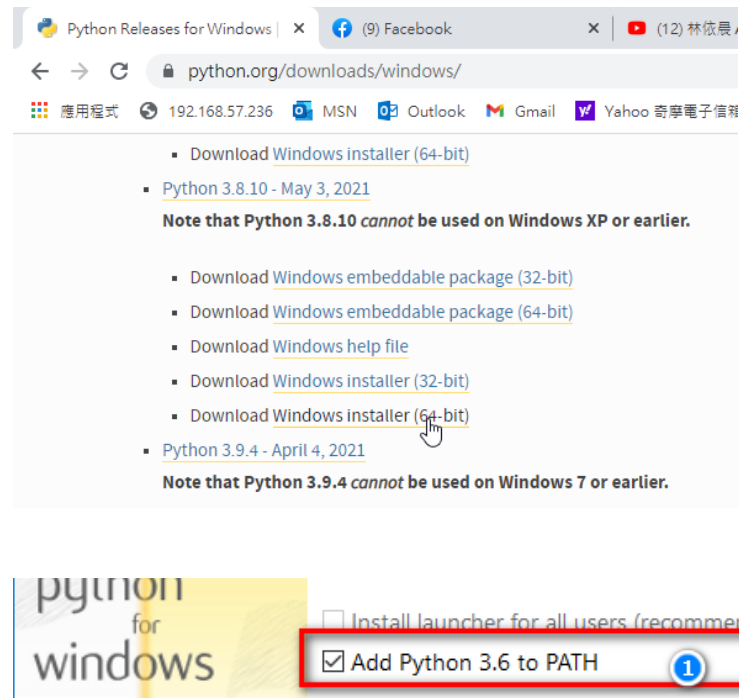


3-1 Django 平台建置(windows)



➤ 若已裝多個 python 版本：

若已裝兩種版本的 python，也新增到環境變數中。

python -version (查看當下版本，Windows PATH 決定)

python -V (查看當下版本，Windows PATH 決定)

如果有安裝多個 Python，可以查看所有版本：

py -0

```
C:\Users\tony>python -V
Python 3.14.0

C:\Users\tony>python --version
Python 3.14.0

C:\Users\tony>py -0
-V:3.14 *      Python 3.14 (64-bit)
-V:3.9         Python 3.9 (64-bit)
```

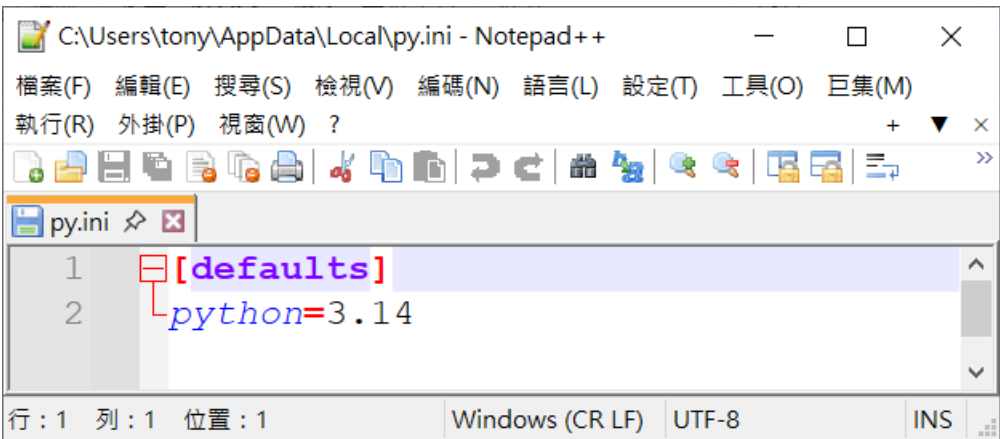
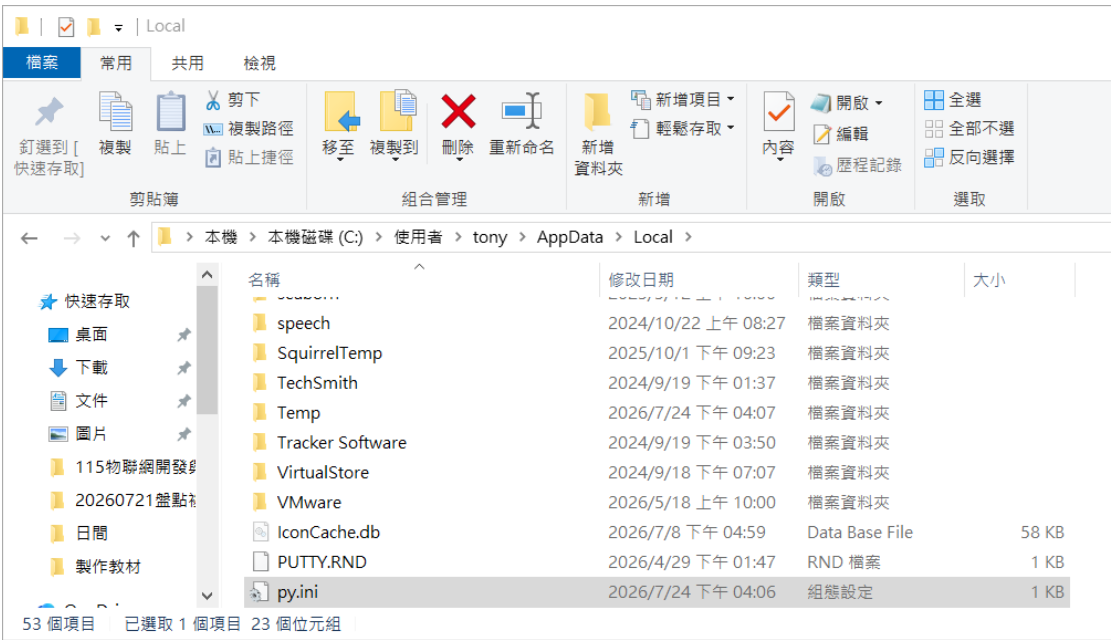
py -V (由 Python Launcher (C:\Windows\py.exe) 控制，可以透過 py.ini 設定預設版本)

方法一：每次指定版本（最簡單，推薦）

py -3.9（每次指定版本）

方法二：修改 Python Launcher 的預設版本

新增文件「C:\Users\tony\AppData\Local\py.ini」



[defaults]

python=3.14

py -V

```
C:\Users\tony>py -V
Python 3.14.0
C:\Users\tony>
```

➤ 利用 venv 建立虛擬環境：

方法一：使用目前預設 Python：

```
# python -m venv venv1
```

方法二（指定 Python 版本）

```
# py -0
```

```
c:\>py -0
-V:3.14 *      Python 3.14 (64-bit)
-V:3.9        Python 3.9 (64-bit)
```

```
# py -3.9 -m venv venv2
```

啟動虛擬環境，venv1

```
# cd c:\venv1
```

```
# Scripts\activate
```

```
C:\Users\tony> cd c:\
c:\>cd venv1
c:\venv1>Scripts\activate.
```

```
(venv1) c:\venv1>python -V
Python 3.14.0
(venv1) c:\venv1>pip list
Package Version
-----
pip      25.2
(venv1) c:\venv1>_
```

deactivate(離開虛擬環境)

啟動虛擬環境，venv2

```
Microsoft Windows [版本 10.0.19045.4894]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。
C:\Users\tony>cd c:\
c:\>cd venv2
c:\venv2>Scripts\activate_
```

```
(venv2) c:\venv2>python -V
Python 3.9.7

(venv2) c:\venv2>pip list
Package      Version
-----
pip          21.2.3
setuptools   57.4.0
WARNING: You are using pip version 21.2.3; however, version 26.0.1 is available.
You should consider upgrading via the 'c:\venv2\Scripts\python.exe -m pip install --upgrade pip' command.
```

deactivate(離開虛擬環境)

開發 django 流程

python -m venv venv1

cd c:\venv1

Scripts\activate

```
C:\Users\tony> cd c:\
c:\>cd venv1
c:\venv1>Scripts\activate
```

```
(venv1) c:\venv1>python -V
Python 3.14.0

(venv1) c:\venv1>pip list
Package Version
-----
pip      25.2

(venv1) c:\venv1>_
```

安裝 Django 套件：

```
# pip3 install Django
```

```
# pip3 list
```

```
(venv1) c:\venv1>pip list
Package Version
-----
asgiref    3.12.1
Django     6.0.7
pip        25.2
sqlparse   0.5.5
tzdata     2026.3
```

建立專案：

```
# Django-admin startproject project1
```

建立應用程式：

```
# cd project1
```

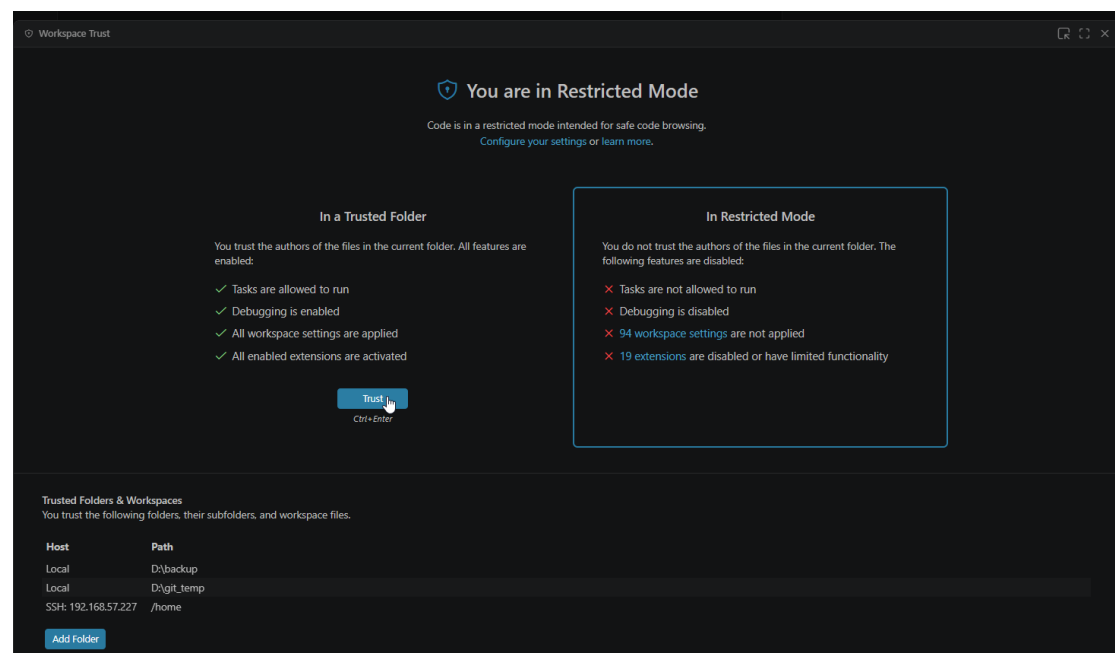
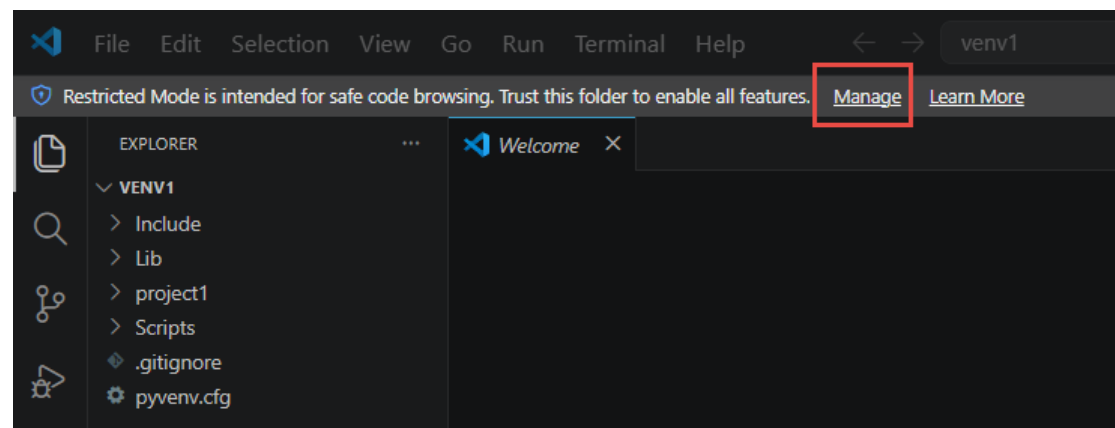
```
# python manage.py startapp myapp
```

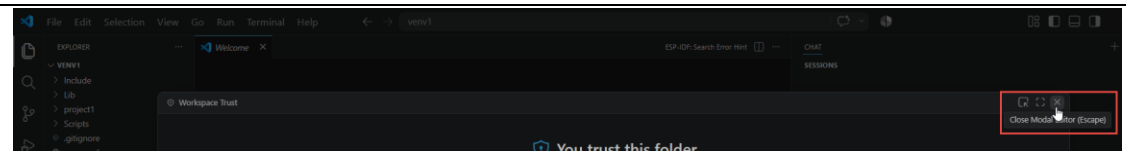
```
# mkdir templates
```

```
# mkdir static
```

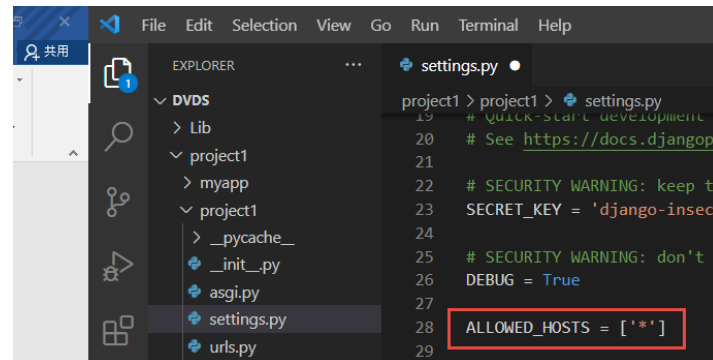
```
(venv1) c:\venv1>Django-admin startproject project1
(venv1) c:\venv1>cd project1
(venv1) c:\venv1\project1>python manage.py startapp myapp
(venv1) c:\venv1\project1>mkdir templates
(venv1) c:\venv1\project1>mkdir static
```

利用 vscode 開啟與修改 settings.py

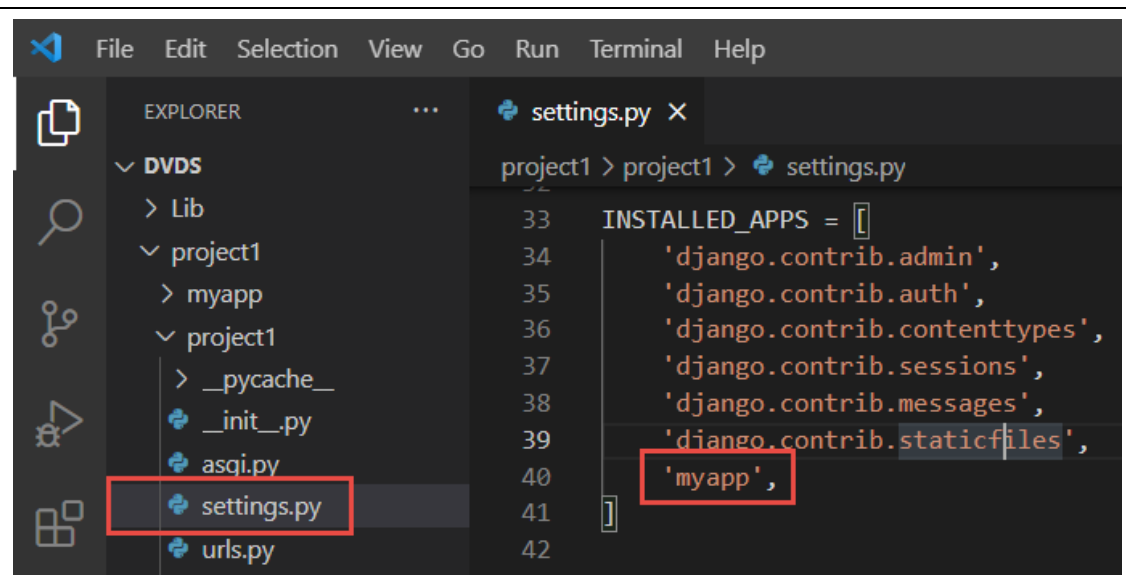




```
ALLOWED_HOSTS = ['*']
```



```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'myapp',  
]
```



```
TEMPLATES = [  
    {  
        'BACKEND': 'django.template.backends.django.DjangoTemplates',  
        'DIRS': [BASE_DIR / 'templates'],  
        'APP_DIRS': True,  
        'OPTIONS': {  
            'context_processors': [  
                'django.template.context_processors.debug',  
                'django.template.context_processors.request',  
                'django.contrib.auth.context_processors.auth',  
                'django.contrib.messages.context_processors.messages',  
            ],  
        },  
    ],  
],
```

```

53  ROOT_URLCONF = 'project1.urls'
54
55  TEMPLATES = [
56      {
57          'BACKEND': 'django.template.backends.django.DjangoTemplates',
58          'DIRS': [BASE_DIR / 'templates'],
59          'APP_DIRS': True,
60          'OPTIONS': {
61              'context_processors': [
62                  'django.template.context_processors.debug',
63                  'django.template.context_processors.request',
64                  'django.contrib.auth.context_processors.auth',
65                  'django.contrib.messages.context_processors.messages',
66              ],
67          },
68      ],
69  ]

```

LANGUAGE_CODE = 'zh-Hant'

TIME_ZONE = 'Asia/Taipei'

```

107  LANGUAGE_CODE = 'zh-Hant'
108
109  TIME_ZONE = 'Asia/Taipei'
110
111  USE_I18N = True
112
113  USE_TZ = True

```

STATIC_URL = 'static/'

```

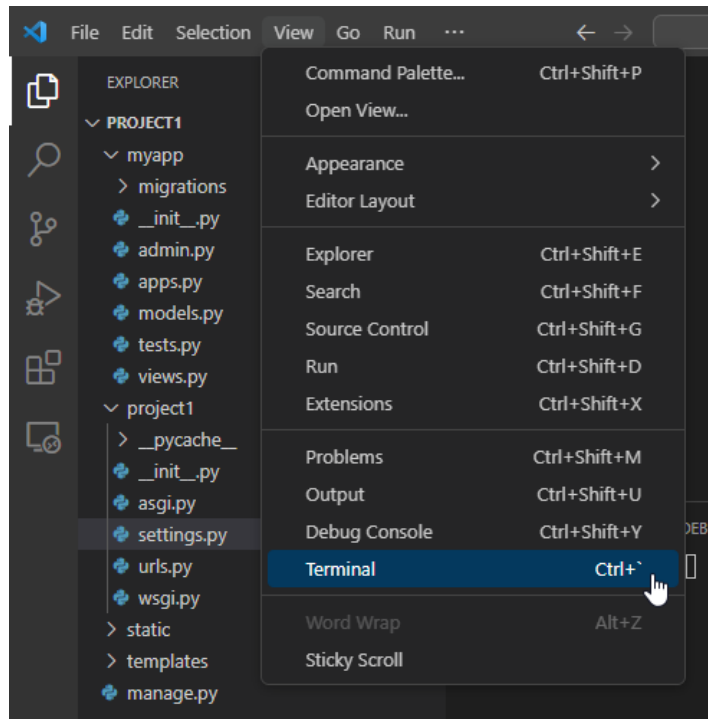
STATICFILES_DIRS = [
    BASE_DIR / 'static',
]

```

```

116 # Static files (CSS, JavaScript, Images)
117 # https://docs.djangoproject.com/en/4.1/howto/static-files
118
119 STATIC_URL = 'static/'
120 STATICFILES_DIRS = [
121     BASE_DIR / 'static',
122 ]
123

```



以系統管理身分執行(PowerShell or VScode)



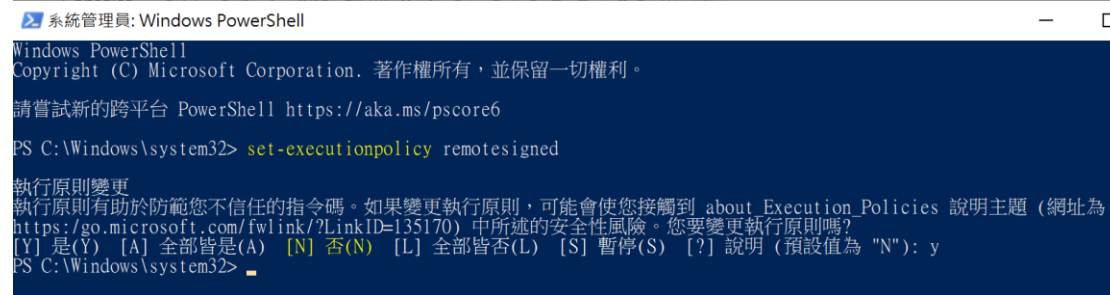
```
# set-executionpolicy remotesigned
```

```
# get-executionpolicy -list
```

說明:PowerShell 更改執行原則

Reference:

https://learn.microsoft.com/zh-tw/powershell/module/microsoft.powershell.core/about/about_execution_policies?view=powershell-7.4



```
系統管理員: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. 著作權所有，並保留一切權利。
請嘗試新的跨平台 PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> set-executionpolicy remotesigned
執行原則變更
執行原則有助於防範您不信任的指令碼。如果變更執行原則，可能會使您接觸到 about Execution_Policies 說明主題（網址為
https://go.microsoft.com/fwlink/?LinkID=135170）中所述的安全性風險。您要變更執行原則嗎？
[Y] 是(Y) [A] 全部皆是(A) [N] 否(N) [L] 全部皆否(L) [S] 暫停(S) [?] 說明（預設值為 "N"）: y
PS C:\Windows\system32>
```

python manage.py makemigrations myapp (建立 migration 資料檔)

python manage.py migrate (模型與資料庫同步)

```
(dvds) c:\dvds\project1>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
```

or

```
▼ TERMINAL
● PS C:\venv1\project1> ..\Scripts\activate
● (venv1) PS C:\venv1\project1> python manage.py makemigrations myapp
No changes detected in app 'myapp'
● (venv1) PS C:\venv1\project1> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
❖ (venv1) PS C:\venv1\project1> 
```

#python manage.py runserver 0.0.0.0:8080

```
(dvds) c:\dvds\project1>python manage.py runserver 0.0.0.0:8080
Watching for file changes with StatReloader
Performing system checks...

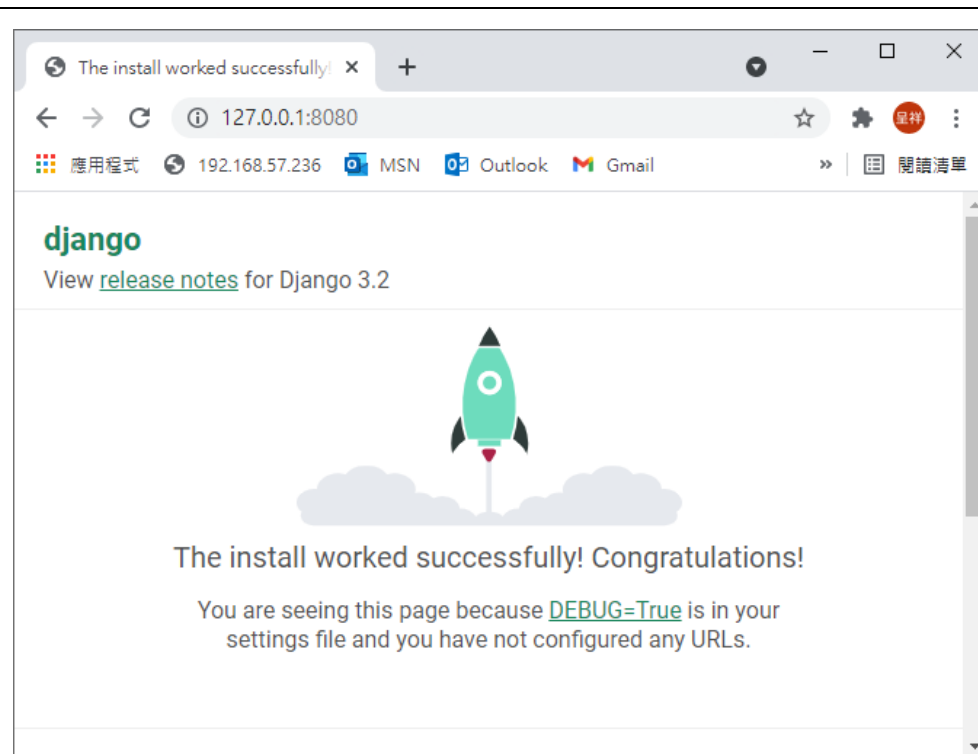
System check identified no issues (0 silenced).
June 11, 2021 - 16:01:42
Django version 3.2.4, using settings 'project1.settings'
Starting development server at http://0.0.0.0:8080/
Quit the server with CTRL-BREAK.
```

Or

```
● (venv1) PS C:\venv1\project1> python manage.py runserver 0.0.0.0:8080
Watching for file changes with StatReloader
Performing system checks...

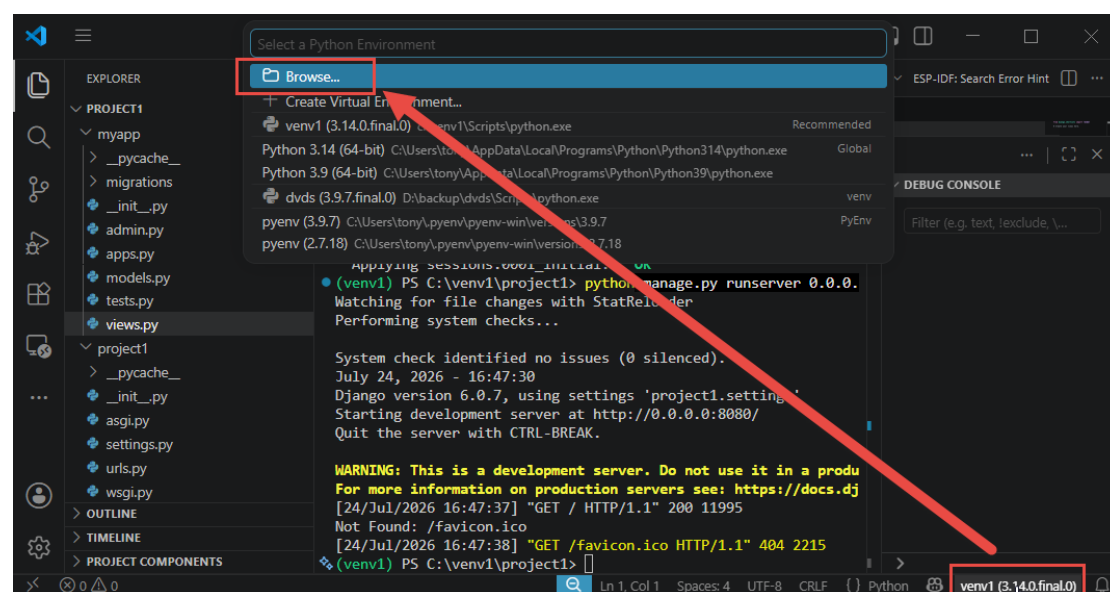
System check identified no issues (0 silenced).
July 24, 2026 - 16:47:30
Django version 6.0.7, using settings 'project1.settings'
Starting development server at http://0.0.0.0:8080/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/6.0/howto/deployment/
[24/Jul/2026 16:47:37] "GET / HTTP/1.1" 200 11995
```



設定 vscode 要啟用那個虛擬環境：

方法一(手動)：



Select Python Executable

← → ↕ ↑

本機 > 本機磁碟 (C:) > venv1 > Scripts

組合管理 ▾ 新增資料夾

115物聯網開發
日間
汽機車資料
製作教材
OneDrive
本機
3D 物件

名稱	修改日期	類型
django-admin.exe	2026/7/24 下午 04:37	應用程式
pip.exe	2026/7/24 下午 04:28	應用程式
pip3.14.exe	2026/7/24 下午 04:28	應用程式
pip3.exe	2026/7/24 下午 04:28	應用程式
python.exe	2025/10/7 上午 11:17	應用程式
pythonw.exe	2025/10/7 上午 11:17	應用程式
sqlformat.exe	2026/7/24 下午 04:37	應用程式

project1

EXPLORER

PROJECT1
myapp
 __pycache__
 migrations
 __init__.py
 admin.py
 apps.py
 models.py
 tests.py
 views.py
project1
 __pycache__
 __init__.py
 asgi.py
 settings.py
 urls.py
 wsgi.py
OUTLINE
TIMELINE
PROJECT COMPONENTS

views.py

myapp > views.py
from django.shortcuts import render

TERMINAL

(venv1) PS C:\venv1\project1> python manage.py migrate
Applying auth.0011_update_proxy_permissions... OK
Applying auth.0012_alter_user_first_name_max_length... OK
Applying sessions.0001_initial... OK
(venv1) PS C:\venv1\project1> python manage.py runserver 0.0.0.
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
July 24, 2026 - 16:47:30
Django version 6.0.7, using settings 'project1.settings'
Starting development server at http://0.0.0.0:8080/
Quit the server with CTRL-BREAK.

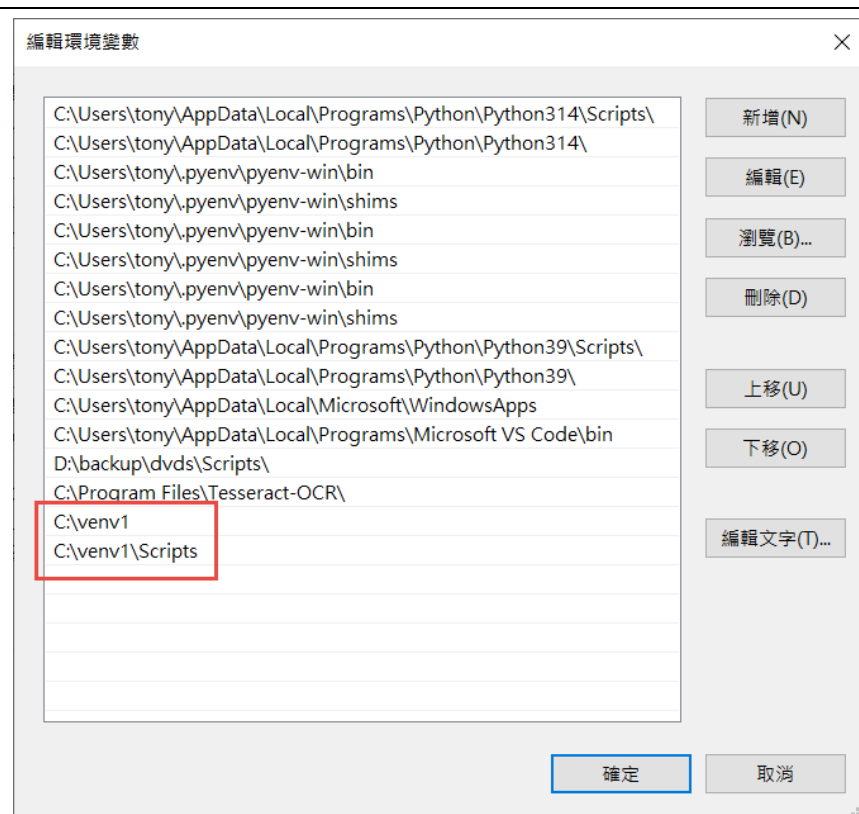
WARNING: This is a development server. Do not use it in a produ
For more information on production servers see: https://docs.dj
[24/Jul/2026 16:47:37] "GET / HTTP/1.1" 200 11995
Not Found: /favicon.ico
[24/Jul/2026 16:47:38] "GET /favicon.ico HTTP/1.1" 404 2215
(venv1) PS C:\venv1\project1>

DEBUG CONSOLE

Filter (e.g. text, exclude, \...

Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Python venv1 (3.14.0.final.0)

方法二(新增環境變數):



新增組態（目的讓 vscode 知道使用什麼平台）：

